

Compton scattering

Shi Teng Mei

Fall 2024

Abstract: This experiment, conducted at Stony Brook University as part of the senior lab course in December 2024, investigates the principles of the Compton effect to verify the rest energy of an electron and the energy of an incident gamma ray. The measured rest energy of the electron was approximately 0.570 ± 0.016 MeV, and the energy of the incident gamma ray was approximately 0.705 ± 0.017 MeV. The experimental setup includes a Compton scattering apparatus designed to measure the energy and count rate of scattered gamma rays within a specific solid angle.

Using the collected data, we constructed a graph of the differential cross section and compared it to theoretical predictions from the Thomson and Klein-Nishina (K-N) formulas. The calculated minimum chi-square to K-N and Thomson model of differential cross section are 70.16 and 1340.37, respectively, indicating that the K-N model provides a better approximation for the observed experimental results. Additionally, we determined the number of total free electrons in small aluminum (Al) and copper (Cu) scatterers and analyzed their yield ratios at scattering angles of 25° and 45° , obtaining values of 0.86 and 0.64, respectively.

I. Introduction

The Compton scattering experiment is one of the popular experiments that supports the wave-particle duality of light. In 1920, Arthur Holly Compton (A. H. Compton) observed that after X-rays are scattered by free electrons in the scatterer the energy of the X-rays had decreased, and he also found that the wavelength of the scattered wave is slightly larger than the wavelength of the incident ray, and this is known as the Compton effect. The Compton shift $\Delta\lambda$ represents the change in wavelength. This is due to the incoming photon colliding with an electron and sharing some of its energy with it, so the scattered photon has less energy than before and so the wavelength of the scattered photon is larger.

During the scattering, some of the energy and momentum of the photons are transferred to the electrons, much like a collision between particles. This shows the observation of light behaving as if it were a particle or called a photon. The Compton scattering experiments prove the particle-wave duality by showing that these results are consistent with the Compton effect formula, in which this formula is derived based on the fundamental principle of conservation of momentum and energy in each photon-particle collision and Einstein's theory of relativity. The result of the experiment was consistent with the Compton effect formula, and this confirmed that light (photons) has both energy and momentum conserved during the interaction between photons and matter, which was essential for the development of quantum theory.

Also it is important to notice that since the momentum is conserved, the energy shift of the photons depends on the scattering angle, and the formula of Compton effect will also depend on the scattering angle.

These incident rays are scattered into all directions around the scatterer, and the probability density of observing scattered rays in a particular direction per unit of solid angle depends on the scattering angle. This scattering probability, also known as the differential scattering cross-section ($d\sigma/d\Omega$), is well described by the Klein-Nishina (K-N) formula, developed in 1929 by Oskar Klein and Yoshio Nishina.

Furthermore, we expect that the scattering cross section as a function of angle agrees with the predictions of quantum field theory (Klein-Nishina formula) rather than with classical Thomson scattering. In Thomson scattering, the incident light wave hits a free electron, it causes the electron to oscillate at the same frequency, emitting a delayed electromagnetic radiation in response. This re-radiation is what we observe as scattered light. The photon energy remains unchanged since the process is elastic. The photon simply changes direction without transferring any energy to the electron, which fits with a classical view of wave interaction. Unlike Compton scattering, which requires a quantum-mechanical treatment to account for energy transfer between photons and electrons.

Thomson scattering works well within classical mechanics. It applies when the energy of the photons is low relative to the electron rest mass energy (i.e., much lower than 511 keV), and relativistic effects are negligible. Hence, the incident wave experiences a change in momentum direction (not magnitude) as it's scattered by the electron. The electron may gain negligible translational momentum in response to the change in momentum direction of the scattered photon. However, since Thomson scattering occurs with low-energy (non-relativistic) photons, the recoil momentum imparted to the electron is minimal and does not significantly alter its kinetic energy.

II. Compton scattering setup description:

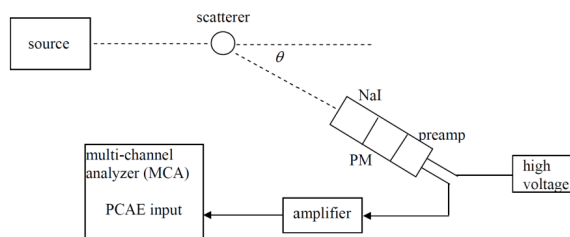


Fig 1. The figure above is an excerpt from the lab-manual of this course. It is a schematic of the Compton scattering experiment. You measure the scattering of a strong ^{137}Cs source which emits γ -rays with a single energy of 661.6 keV. Notice, the source is heavily and permanently shielded in a lead housing. (You must always use gloves when handling lead, because its oxide is poisonous.)

- The setup of Compton scattering

This experiment was performed at Stony Brook University in fall 2024 under the supervision of professor Jan C. Bernauer. The components of the experimental setup include a setup of Compton scattering shown in figure 1, and a large aluminum scatterer, a small Al, and a small Cu scatterer. Also a few weaker sources: ^{137}Cs , ^{22}Na , ^{133}Ba , and ^{60}Co .

We want to verify the Compton effect formula that has been derived and as well as to observe how well the K-N equation describes the experimental data that will be collected. First, notice that shown in figure 1 the radiation source emits gamma rays, which are detected by the NaI crystal and scintillate (emit visible light). The gamma rays excite the electrons in the NaI crystal and as these excited atoms return to their lower-energy state with a decay time constant of

$\tau=850$ ns, they release the absorbed energy as lower-energy visible light.

Second, the scintillation light is detected by a photomultiplier tube (PMT) attached to the crystal. The PMT converts the light pulses into electrical pulses, which are proportional to the energy of the incoming gamma photons.

Third, the weak electrical pulses from the PMT are then fed into a preamplifier. The preamplifier boosts the signal without significantly altering its shape. It provides initial amplification to ensure that the signal is strong enough for further processing.

Fourth, use an oscilloscope to examine the shape of the pulses directly from the preamplifier. These pulses should reflect the exponential decay shape of the scintillation light with a time constant of 850 ns. The pulse height (amplitude) shown on the oscilloscope is directly proportional to the energy of the gamma photons detected by the NaI crystal.

Finally, the output from the shaping amplifier is sent to a computer with a Multi-Channel Analyzer (MCA) program. The MCA digitizes the pulse heights and sorts them into channels, creating a pulse height spectrum. Each channel corresponds to a specific energy range, allowing you to analyze the energy distribution of the incoming gamma rays.

In a pure Compton scattering experiment, you will only observe the interaction of gamma radiation with electrons, leading to scattered photons and possibly ejected electrons. However, in this experiment there is β^+ decay or positron emission involved in this process as you will see in the radiation of ^{22}Na .

The detector in this experiment is a 3×3 in² in cylindrical NaI(Tl) crystal. In theory we can increase the probability for gamma rays interacting fully within the detector before escaping by using a larger detector since it captures more scattered photons. Hence, the photopeak becomes more distinct because More gamma rays deposit their full energy in the detector and fewer gamma rays escape after partial energy deposition (reduction in the Compton contribution).

- Goals of the experiment

To verify the Compton effect formula, we do a simple trick, which is we can take the inverted of the compton effect formula, so its y-value is the $1/E'$, and its slope

is $\frac{1}{m_e c^2}$ with $1 - \cos(\theta)$ as the dependent variable, and $1/E$ as the y-intercept. Obviously, if we can find the energies of the scattered rays at specific scattering angles. Then the line of best fit must be equal to the inverted rest-mass-energy of an electron, and the y-intercept would be the value of the inverted energy of the incident gamma ray.

$$y = Ax + B \Rightarrow \frac{1}{E'} = \frac{1}{m_e c^2} (1 - \cos(\theta)) + \frac{1}{E} \quad (1)$$

To verify the differential cross section equation, we need to compare the graph of the experimental differential cross section vs theta to the K-N and Thomson model depending on the scattering angle.

$$Y(\theta) = N_e N_\gamma \frac{d\sigma}{d\Omega}_{\text{experimental}} \Rightarrow \frac{d\sigma}{d\Omega}_{\text{experimental}} = \frac{Y(\theta)}{N_e N_\gamma d\Omega} \quad (2)$$

III. Theory

A. Derivation of the Compton effect equation

In this lab, the Compton scattering experiment, we replace the X-rays with nuclear γ -rays. As the incident photons of energy $E_\gamma = h\nu$ are deflected by a free (stationary) electron, energy conservation and momentum conservation requires that the photon energy is decreased to $E'_\gamma = h\nu'$ while the recoiling electron acquires an energy $E_e = E_\gamma - E'_\gamma$.

When a particle moves close to the speed of light ($v \approx c$), its momentum contributes significantly to its total energy. In this regime, the kinetic energy is no longer well-approximated by the classical formula ($KE = \frac{1}{2}mv^2$). Instead,

$$E = \sqrt{(mc^2)^2 + (pc)^2}, \quad \text{Velocity}_{\text{relativistic}} = \frac{dE}{dp} = \frac{pc^2}{\sqrt{(mc^2)^2 + (pc)^2}} \quad (3)$$

In this experiment, we use high energy gamma rays produced by the sources, such as ^{137}Cs , ^{22}Na , ^{133}Ba , and ^{60}Co . (The "22" in ^{22}Na represents the mass number of the sodium isotope) Since they are electromagnetic rays, they are massless. Therefore, $E_{\text{photon}} = pc = h\nu$, where $p = \frac{h\nu}{c} = \frac{h}{\lambda}$, and $\frac{dE}{dp}_{\text{photon}} = c \quad (4)$

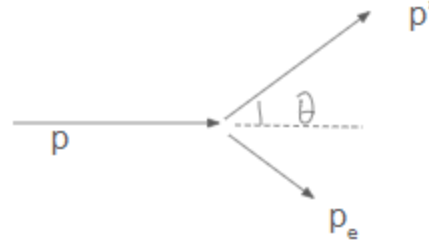


Fig 2. This is a picture of Compton scattering. The incident ray interacts with a scatterer, producing a scattered ray while also ejecting some free electrons.

Where p , p' , and p_e are the momentum vectors of the initial gamma ray, final gamma ray, and the electron.

Conservation of momentum: $p = p' + p_e \Rightarrow p_e^2 = (p - p')^2 = p^2 + p'^2 - 2p \cdot p'$

Conservation of energy: $E_i = pc + m_e c^2 = h\nu + m_e c^2$,

$$E_f = p'c + \sqrt{(m_e c^2)^2 + (p_e c)^2} = h\nu' + \sqrt{(m_e c^2)^2 + (p_e c)^2} \quad (5)$$

The use of conservation of energy and momentum and the Planck-Einstein relation can derive the formula of the Compton effect., which is shown below.

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos\theta) \Rightarrow E'_\gamma = E_\gamma / [1 + \frac{E_\gamma}{m_e c^2} (1 - \cos\theta)] \quad (6)$$

Since $E = h\nu$ We can derive the E_γ by further replace $\lambda = \frac{c}{\nu} = \frac{hc}{E}$, and then isolate E' .

B. The K-N differential cross section for scattering:

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2}{2} \times \frac{1 + \cos^2\theta}{[1 + \gamma(1 - \cos\theta)]^2} \times \left\{ 1 + \frac{\gamma^2(1 - \cos\theta)^2}{(1 + \cos^2\theta)[1 + \gamma(1 - \cos\theta)]} \right\} \quad (7)$$

Where $r_0 = 2.82 \times 10^{-13}$ cm is the radius of the electron and $\gamma = \frac{E_\gamma}{m_e c^2}$. The equation tells you the probability of particles scattering into a specific direction per unit solid angle.

Obviously, when $E_\gamma \ll m_e c^2$ nonrelativistic then γ become negligible, so the K-N equation reduce to the Thomson formula (classical formula)

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2}{2} \times (1 + \cos^2\theta) \quad (8)$$

Planck's constant h is absent from the K-N formula because the h , is hidden within the photon energy $E_\gamma = h\nu$.

C. How does a photomultiplier tube (PMT) work?

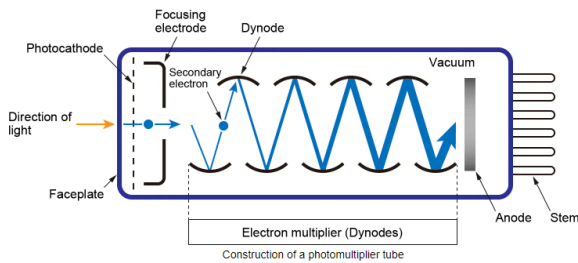


Fig 3. This figure is taken from the website [1]. Photomultiplier tube (PMT) is a highly sensitive device used to detect and amplify low levels of light by using the photoelectric effect and electron multiplication.

When light (photons) enters the PMT, it strikes the photocathode - a photosensitive material with a low work function. Through the photoelectric effect, each incident photon can eject an electron from the photocathode. The number of ejected electrons is proportional to the light intensity, so low-intensity light yields only a few electrons. The ejected electrons are accelerated (through an electric field produced by dynodes) toward a series of dynodes. Each dynode is held at a progressively higher positive voltage relative to the one before it.

When the electrons strike the first dynode, they release multiple secondary electrons. These new electrons are then directed to the next dynode. This process repeats at each dynode stage, leading to an avalanche multiplication effect. A typical PMT has 10 to 12 dynodes, allowing amplification factors of up to 10^6 to 10^7 .

After passing through all the dynodes, the multiplied electron cloud reaches the anode. At the anode, the large number of electrons creates a measurable electric current, which is proportional to the intensity of the original light signal. This current is then further processed as a signal that can be measured and analyzed.

D. Distinction between the photopeak, Compton peak, backscatter peak, Compton edge, Compton continuum.

Since a photopeak is a peak that represents the energy of the gamma rays fully absorbed by the detector. So in the calibration, we measure the channel number corresponding to the photopeak of each source scattered without a scatterer. Whenever rays are not heading straight forward to the detector,

these rays lose some of their energy by scattering off materials outside the detector (like shielding, the environment, or the source itself) before reaching the detector. That's why we don't care about peaks like the backscatter peak, Compton edges (lower energy peaks appearing to the left of the photopeak.), and Compton continuum since they have lost some energy before reaching the detector. Those rays heading straight towards the detector are supposed to be the ones that contribute to the photopeak of the source of radiation.

When we measure the energy of the scattered ray during a Compton scattering, we again consider the peak with the highest energy level compared to rest of peaks of the spectrum of the scattering as the energy of the scattered ray, where this peak is called the Compton peak for a given scattering angle. Scattered rays at other angles contribute to the Compton continuum or other spectral features (e.g., backscatter peak), but not to this peak.

Moreover, it is important to notice that the energy of a scattering ray follows the eq (6), these scattering lights have the scattered energy depending on the scattering angle.

IV. Experiment design

A. Calibration:

Since what we will be measuring and seeing on the computer screen is a bunch of counts of scattered rays shown in a graph of histogram (counts v.s. Channel numbers). Hence to measure the energy of the scattered rays we need to calibrate these channel numbers to specific energy levels before we can analyze any measurements. To do so, we need to start with a few weaker sources of radiation (^{137}Cs , ^{22}Na , ^{133}Ba , and ^{60}Co) without scatterer and measure the photopeak of the spectrum and these peaks are associate with known full energy of the corresponding incident rays, and we want to subtract a spectrum of incident radiation from the background (without any sources exposed to it) since we don't want any other background radiation that may affect the energy of the incident gamma rays. With measured channel number and the corresponding full-energy of the incident gamma rays of each of these radiation sources, we now can calibrate the channel number to specific energy levels.

To measure the channel number of the photopeak we should make a gaussian distribution curve over the photopeak and the mean value of the fitting gaussian curve will be the channel number of the photopeak for the source of radiation.

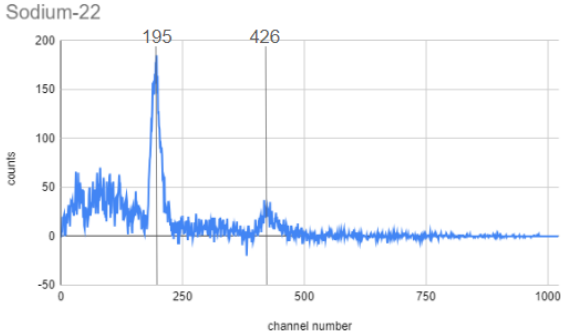


Fig 4. For the ^{22}Na , the peak at about 1.26 MeV arises from the de-excitation of ^{22}Na ; the larger peak at about 0.54 MeV arises from annihilation radiation. Indeed, from the level diagram of ^{22}Na decay, we notice that positrons are emitted, so this experiment is not a pure Compton scattering.

radiation source	Channel #	sigma_channel	photopeak (MeV)
sodium-22	426	17	1.275
cesium-137 (weak)	244	11	0.662
barium-133 (weaker transition)	38	7	0.081
barium-133 (stronger transition)	137	17	0.356
cobalt-60 (weaker transition)	393	11	1.173
cobalt-60 (stronger transition)	442	11	1.333

Table 1. Calibrate channel number to specific energy level

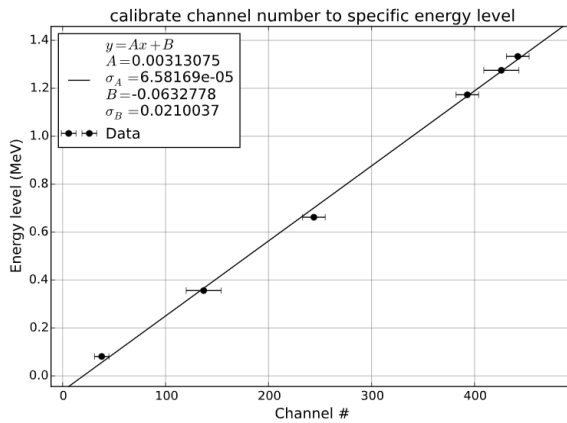


Fig 5. These pairs of raw data were collected from four different weaker sources [table 1], but we have six pairs of data because some sources, such as ^{60}Co and ^{133}Ba each have two distinct photopeaks.

$$E = (0.0031 \pm 0.0001)\text{channel} + (-0.0633 \pm 0.0210),$$

$$\sigma_E = \sqrt{(\text{channel} \cdot \sigma_B)^2 + (B \cdot \sigma_x)^2 + \sigma_c^2} \quad (9)$$

B. Verify Compton effect formula

We can now focus on a single radiation source and verify the eq (6), (7), and (8). As I mentioned that eq (1) has its y-value as the $1/E'$, and its slope is $\frac{1}{m_e c^2}$

with $1 - \cos(\theta)$ as the dependent variable, and $1/E$ as the y-intercept. That is, we can measure the energy of the scattered rays at some scattering angles, so we can form a linear regression line, and verify if the slope and the y-intercept values are consistent with eq (1). First we place the large Al scatterer at the pivot point. Open the exit hole of the strong source ^{137}Cs . Because the gamma rays may exit in all directions, it is possible for the detector to catch up to some of the unscattered gamma rays directly without scattered by the scatterer, so we need to arrange lead shielding (those lead blocks) to prevent direct gamma rays from the source from reaching the detector (the crystal).

Next, we will measure the position of each photopeak by using the gaussian fitting curve and compute the energy of the scattered γ -ray, E' , for each of the Compton peaks that we found at each of the scattering angles by using the linear relations channel-to-energy (Fig 5).

C. Verify differential cross section by comparing to the experimental differential cross section.

We can examine differential cross section eq (7) and (8) by comparing to our experimental differential cross section computed in eq (10) shown below. We start at about 18 degrees because it is hard to separate the incoming beam from the scattered beam at smaller angles. We have taken measurements at 18°, 20°, 25°, 30°, 40°, 50°, etc. until about 110°.

I mentioned that eq (2), $Y(\theta) = N_e N_\gamma \frac{d\sigma}{d\Omega}_{\text{experimental}} \Rightarrow \frac{d\sigma}{d\Omega}_{\text{experimental}} = \frac{Y(\theta)}{N_e N_\gamma d\Omega}$

$$N_e (\text{total number of electrons in the target}) = \pi \left(\frac{d}{2}\right)^2 h \rho \frac{N_0}{A} Z \quad (10)$$

$$N_\gamma (\text{flux of incident } \gamma \text{ rays per unit area at } 0^\circ \text{ without scatterer}) = \frac{(\text{total incident rays})}{d \cdot h} \quad (11)$$

$$d (\text{an equivalent square thickness of the scatterer}), I_{\text{trans}} = I_{\text{incident}} e^{-\mu \rho d} \quad (12)$$

$$h (\text{height of the target}), \rho (\text{density of target}), N_0 (\text{Avogadro's number}) = 6 \cdot 10^{23}$$

$$A (\text{atomic weight of atoms of the target}), Z (\text{atomic number of atoms of the target})$$

$$d\Omega (\text{solid angle subtended from the scatterer to the detector}) = \frac{\text{crystal area}}{r^2} = \frac{7.62 \text{ cm}^2}{28.5 \pm 0.1 \text{ cm}^2} = 0.0715 \pm 0.0143 \quad (13)$$

N_γ is the flux of incident γ -rays per unit area hitting the scatterer during the data run. We can figure this out by detecting the flux with the detector at 0° without the scatterer in place, in which the counts under the entire spectrum is the flux of incident γ -rays. Where d is the thickness calculated from the eq (14) shown below.

To determine $Y(\theta)$, you should integrate the area under the curve of the full-energy peak (or Compton peak) to get the total number of counts under the full-energy peak with an assigned uncertainty, after

subtracting any background. This will give us a more accurate measure of the scattered gamma rays detected at that angle because of the finite energy resolution of the detector.

With the scatterer in place the beam is attenuated. You can measure the attenuation by placing the scatterer in place and repeating the run. Each time integrate over the full energy peak. The attenuation is described by a mass attenuation coefficient, $\mu(E_\gamma)$, depending on E_γ the energy of the incident gamma ray, sometimes called mass absorption coefficient.

$$I_{\text{trans}} = I_{\text{incident}} e^{-\mu \rho d} \Rightarrow d_{\text{experimental}} = \frac{1}{\mu \rho} \ln\left(\frac{I_{\text{incident}}}{I_{\text{trans}}}\right) = \frac{1}{\mu \rho} \ln\left(\frac{\text{total incident rays at } 0^\circ \text{ no scatterer}}{\text{Total scattered rays at } 0^\circ \text{ with scatterer}}\right) \quad (14)$$

μ is given in the appendix for Al, $\mu(E_\gamma) = 0.074 [\text{cm}^2/\text{g}]$ at $E_\gamma = 661.6 \text{ keV}$

$$\rho (\text{density of aluminum}) = 2.7 \text{ g/cm}^3. \quad d_{\text{experimental}} = 4.98 \pm 0.34 \text{ cm}$$

Since that d is the effective thickness of the scatterer, we can check to see if that d value we obtained makes sense by comparing it to an estimated average thickness of the target.

$$d_{\text{avg}} = \frac{\pi r^2}{d(\text{direct measure})} = 4.99 \text{ cm with } d_{\text{meas}} = 6.35 \pm 0.1 \text{ cm}$$

There's about 2% difference between the experimental thickness of the scatterer and the average thickness of the target, so I believe that this equivalent square thickness of the scatterer makes sense at least in this order.

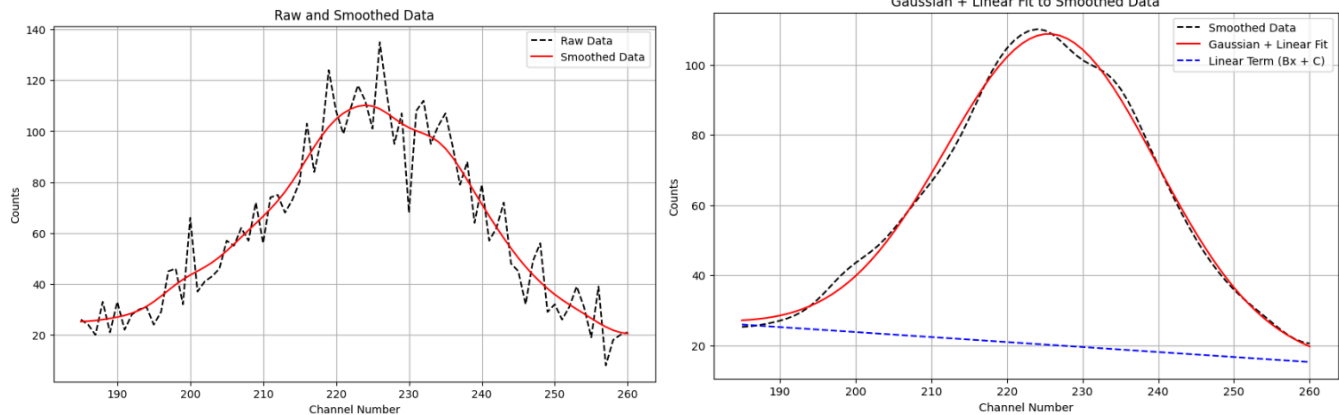


Fig 6. This is a Compton peak portion of the entire Compton scattering spectrum. Since it's hard to make a gaussian curve over these noisy raw data points. I try to smooth the curve first, then do the gaussian fitting curve with a Compton continuum cut by adding $bx+c$. Then we can compute the channel number corresponding to the peak and the total number of counts under the gaussian curve and above the linear fitting line.

D. Yield ratio of two distincted scatterers

We know that the experimental result for the differential cross section is slightly different from the prediction by the K-N and the Thomson model. This is because we have assumed that all electrons in the scatterer are free. However, in real experiments, electrons are bound to atoms, and different binding energy due to different scatterers affect the scattering process. This is particularly significant for low-energy photons, where atomic binding effects cannot be ignored. Moreover, electron density varies from material to another material.

Thus, it is essential to further verify our result. We should note that the differential cross section is just the probability density of observing scattered rays in a particular direction per unit of solid angle depending on the scattering angle but not the material itself.

$$\text{So, } N_e = Y(\theta) / (N_y \frac{d\sigma}{d\Omega} d\Omega) \Rightarrow \text{ratio} = \frac{N_{e,Cu}}{N_{e,Al}} = \frac{Y(\theta)_{Cu}}{Y(\theta)_{Al}} \quad (15)$$

We simply make a comparison of cross section measurements at 25° and 45° by comparing the full-energy-peak areas of the photopeaks for the two scatterers (small Al and small Cu) at angles (25 and 45 degrees) and form the ratio. This detected number of scattered rays corresponds to the free electrons in targets.

scatterer	theta (degrees)	Y(theta)
small Aluminum	25	2090
	45	2000
small copper	25	1792
	45	1284

small Cu over small Al	yield ratio
at (25°)	0.86
at (45°)	0.64
average:	0.75

Table 2. Above shows the total number of scattered rays detected by the detector under the Compton peak. The yield ratios at angle 25 and 45 are about 0.86 and 0.64 respectively. In theory, the geometries for the two samples are essentially the same disregarding the scattering angles and will cause no difference in the ratio, I should use the average value between 0.86 and 0.64 as my better yield ratios, that is about 0.75.

So based on eq (15), we expect that there's more free electrons in the small Aluminum scatterer than the free electrons in the small Copper scatterer.

V. Analysis:

A. Verify the formula of Compton effect through the measured rest-mass-energy of an electron.

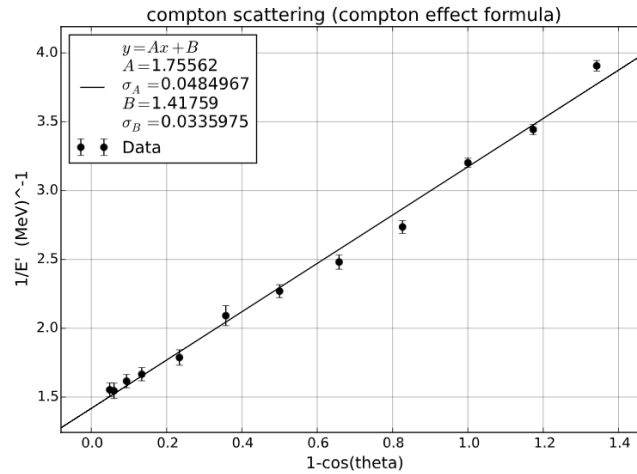


Fig 7. Above is the least-squares fit for $1/E'$, where E' is the energy of the scattered ray depending on scattering angles {18, 20, 25, 30, 40, ...110 degrees}. According to the Compton effect equation the slope of this line of best fit is equal to the rest-mass-energy of an electron. As well as, the y-intercept represents the energy of a single incident gamma ray. $y = Ax + B \Rightarrow \frac{1}{E'} = \frac{1}{m_e c^2} (1 - \cos(\theta)) + \frac{1}{E}$

The minimum chi-squared, $\chi^2_{min} = \sum_{i=1}^n \left(\frac{\text{Observe}_i - \text{Expect}_i}{\sigma_{\text{Expect},i}} \right)^2 \approx 21$. For a P_value ≈ 0.02 or 2%, represents the probability of observing a chi-squared value at least as extreme as the calculated minimum chi-squared of 21), if the null hypothesis is true.

In a 1% significance level test. Since the p-value (2%) is greater than the significance level (1%), we fail to reject the null hypothesis. The null hypothesis, which assumes that the model provides a good fit to the data, and we found out that we cannot be rejected at the 1% significance level. This suggests that the line of best fit (using the model $y=ax+b$ is reasonably reliable for describing the relationship between the $1 - \cos\theta$ and $1/E'$.

While the p-value is relatively low, indicating that the fit may not be perfect, it is not low enough to rule out the null hypothesis at the chosen level of significance.

Rest-mass-energy of electron (E_0) [MeV]	Sigma_E_0 [MeV]	theoretical (E_0) [MeV]	% difference
0.57	0.016	0.511	11.5
Energy of incident y-ray (E) [MeV]	Sigma_E [MeV]	theoretical (E) [MeV]	% difference
0.705	0.017	0.662	6.5

Table 3. The table above shows the rest mass energy of an electron and the energy of an incident gamma ray, and comparing these values to the theoretical values. I found out that the % difference between the experimental and theoretical for E_0 and E is about 11.5% and 6.5% respectively.

B. Comparing the experimental Scattering result with a large aluminum scatterer.

We want to compare the experimental differential cross section to the theoretical K-N and the Thomson model by collect 12 data of scattering from 18 degrees up to 110 degrees, and make a graph of $d\sigma/d\Omega$ versus θ , and we can show which model is more precisely describes the experimental result by calculate the minimum chi-squares with respect to each of the models.

thickness d_experimental(cm)	sigma_d_experimental(cm)	thickness d_measured_avg(cm)	
4.98	0.34	4.99E+00	
N_e (counts)	sigma_N_e (counts)	N_y (rays/m^2)	sigma_N_y (rays/m^2)
2.62E+26	3.58E+25	5208	357

Table 4. We have to calculate the equivalent square thickness of the scatterer, and the number of electrons in the large Al scatterer. And the flux of incident gamma rays hitting the scatterer per unit area during the running time.

theta (degrees)	Y(theta)	$d\sigma/d\Omega$ _Experimental (m^2/sr)	sigma_d $\sigma/d\Omega$ _Experimental (m^2/sr)	$d\sigma/d\Omega$ _K_N (m^2/sr)	$d\sigma/d\Omega$ _Thomson (m^2/sr)
18	3678	3.77E-26	9.49E-27	6.71E-26	7.57E-26
20	3546	3.63E-26	9.14E-27	6.46E-26	7.49E-26
25	3292	3.37E-26	8.48E-27	5.80E-26	7.24E-26
30	2647	2.71E-26	6.82E-27	5.13E-26	6.96E-26
40	2281	2.34E-26	5.89E-27	3.88E-26	6.31E-26
50	1933	1.98E-26	4.98E-27	2.90E-26	5.62E-26
60	1575	1.61E-26	4.05E-27	2.20E-26	4.97E-26
70	1417	1.45E-26	3.65E-27	1.75E-26	4.44E-26
80	1328	1.36E-26	3.42E-27	1.47E-26	4.10E-26
90	985	1.01E-26	2.54E-27	1.31E-26	3.98E-26
100	919	9.42E-27	2.37E-27	1.22E-26	4.10E-26
110	590	6.05E-27	1.52E-27	1.18E-26	4.44E-26

Table 5. This table shows the experimental, K-N, and the Thomson model differential cross section for varying scattering angles from 18 to 110 degrees. As we can see in eq (2), we have assumed that the total number of electrons in the scatterer are free, but in fact not all of the electrons are free, so the experimental differential cross section underestimates the real differential cross section. This is to say, if we use a stronger radiation source then we may get a better result that is better described by the K-N model.

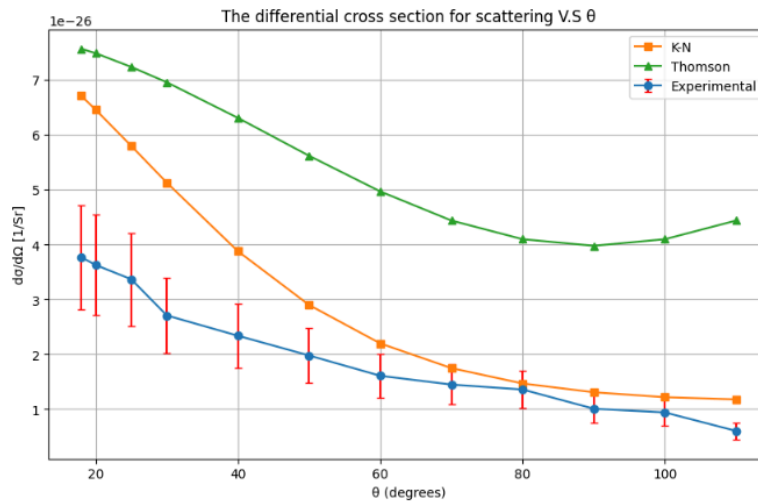


Fig 8. I have graph all three models in a single plot for data in table 5, and I have found that the

$$\chi^2_{min}(\text{to the KN model}) = 70.16 \quad \text{and} \quad \chi^2_{min}(\text{to the Thomson model}) = 1340.37$$

Although the p_values for both of them are about 0 and it is insignificant to test the result in that way, obviously we can still look at the minimum chi-squares to K-N and the Thomson model. Since 70.16 is smaller than 1340.37, it definitely shows that the K-N approximation for the experimentally differential cross section is much better than Thomson approximation. This further prove that the Thomson model may not be effectively on approximate the differential cross section for Compton scattering with a E_γ that doesn't qualify the condition $E_\gamma \ll m_e c^2$.

C. Comparing the Yields of scattered rays scattered by two different scatterers

scatterer	d_meas (cm)	d_expri (cm)	N_e, theory	N_e, experimental
small Al	2.79	2.19	6.91E+25	4.26E+25
small Cu	3.05	2.4	2.69E+26	1.66E+26

Table 6. Experimentally, we found d_expri(small Al) = 2.19 cm and theoretically with d_meas=2.79 cm. There are experimentally about 4.26×10^{25} number of electrons and theoretically about 6.91×10^{25} in small Al. Experimentally 1.66e26 and theoretically 2.69e26 number of electrons in small Cu.

According to eq (10 and 14), I expect that the small Cu scatterer should contain more electrons than the small Al scatterer, shown in table 6.. However, based on table 2, the yield ratios (Cu to Al) at the two angles 25 and 45 degrees are 0.86 and 0.64 respectively, so it implies that there are more free electrons located in the small Al scatterer than in the small Cu scatterer. This contradicts with the ratio of total electrons shown in table 6, where we expect that the small Cu scatterer will have more electrons than the small Al scatterer.

VI. Conclusion:

In this experiment we first calibrate the channel-number to a specific energy level, which is given by $E = (0.0031 \pm 0.0001)\text{channel} + (-0.0633 \pm 0.0210)$. We also compute the rest-mass-energy of an electron and as well as the incident gamma ray by using the formula of the compton effect with the theory that those raw data with $1 - \cos\theta$ as the x-value will have $1/E'$ for the y-value, and $1/m_e c^2$ as the slope and the inverted y-intercept will be our energy of the incident gamma ray. The rest-mass-energy of an electron is about 0.570 ± 0.016 MeV and the energy of the incident gamma ray is about 0.705 ± 0.017 MeV. These results have a 11.6% and 6.5% off from the theoretical or the known value of the rest mass energy and the incident gamma-ray energy. The minimum chi-squared of the line of best fit for the compton effect formula, $\frac{1}{E'} = \frac{1}{m_e c^2} (1 - \cos(\theta)) + \frac{1}{E}$, is about 21 and has a P_value about 2%, so our results for E_0 and E are reliable and accurate with a

fine % off from the known value. That means we have verified the formula of the Compton effect.

The thickness of the large Al scatterer is d_experimental = 4.98 ± 0.34 cm, and the number of electrons in the large Al scatterer is $N_e = 2.62e26 \pm 3.58e25$. The flux of incident γ -rays per unit area hitting the Al scatterer during the data run is $N_\gamma = 5208 \pm 357$. In the graph of differential cross section for scattering v.s. θ our experimental has a min-chi-squared = 70.16 to the K-N model expected by the K-N equation and 1340.37 to the Thomson model. We expect to see that the chi-squared to the K-N model is smaller than the chi-squared to the Thomson formula since the Thomson model is reliable only for non-relativistic $E_\gamma \ll m_e c^2$. Since $70.16 < 1340.37$, we confidently say that the K-N model gives a better approximation for the experimental differential cross section for Compton scattering. Since the experimental differential cross section results are in the same order of magnitude as the K-N and the Thomson model of prediction, we have verified the validity of these models.

In the future, we can prove the reliability of the Thomson model by using a lower source of radiation with the incident gamma ray energy much less than the rest-mass energy of an electron, then we'll expect our experiment differential cross section to v.s. theta has a smaller minimum chi-squared value, however, K-N model is just as good as for both large and small E_γ , since the Thomson equation is just a reduced form of the K-N model, so it could never be better than the K-N model.

Moreover, I found that the yield ratio (Cu over Al) at 25 degrees is 0.86 and 45 degrees is 0.64, but it

contradicts with what we get for the total electrons in the small Al and the small Cu scatterer. The small Cu scatterer is expected to have more electrons by having a total 1.66×10^{26} number of electrons, while the small Al has a total of 4.26×10^{25} number of electrons.

Furthermore, in the future we may try to use a much stronger radiation such that those gamma rays it produces should be stronger enough to free those electrons, and then we may expect to see that the experimental differential cross section result is more fitting to the prediction by the K-N model.

References:

- https://www.matsusada.com/application/ps/photomultiplier_tubes/ [1]
- Text book: Adrian-C.-Melissinos-Jim-Napolitano-Experiments-in-modern-physics